

Surgical Micro Robots For Minimally Invasive Surgery

A Minor Project Report

BACHELOR OF TECHNOLOGY

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COMPUTER SCIENCE & ENGINEERING

Specialization in

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BONAFIDE CERTIFICATE

This is to certify that the minor project titled “Surgical Micro Robots for Minimally Invasive Surgery” has been successfully completed by **Prakhar Gupta (Roll No.: 1230439214)**, **Pradeep Singh (Roll No.: 1230439213)**, and **Rishabh Dwivedi (Roll No.: 1230439240)**, students of the **Bachelor of Technology in the Department of Computer Science & Engineering at Babu Banarasi Das University**, in partial fulfillment of the requirements of the **B.Tech III Year curriculum** during the academic session **2025–26**. The project is a Bonafide work carried out by the students under my supervision. The work presented is original and has not been submitted elsewhere for any academic purpose.

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ABSTRACT

Robotic-assisted surgery has become an important advancement in modern healthcare, providing improved precision, flexibility, and control compared to traditional surgical methods. However, real robotic systems are expensive and not easily accessible for educational and experimental purposes. This project aims to develop a simulation-based solution to understand and demonstrate the working of surgical robots in a controlled virtual environment.

The objective of this work is to design and implement a 3D simulation of a surgical robot operating room using MATLAB. The system includes a patient model placed on an operation table and multiple robotic arms positioned around the patient. The methodology involves the use of MATLAB's 3D graphical functions such as surface plotting and mesh generation to create realistic models. Keyboard-based interaction is implemented to control the movement, rotation, and height of robotic arms in real time.

The results demonstrate a functional and interactive simulation where multiple robots can be controlled smoothly within a 3D environment. The system successfully visualizes robotic movement and coordination around a surgical area. Although the project does not include automation or artificial intelligence, it provides a strong foundation for understanding robotic systems and can be extended further with advanced features such as AI-based control and collision detection.

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LIST OF ABBREVIATION

AI	–	Artificial Intelligence
ML	–	Machine Learning
MATLAB	–	Matrix Laboratory
3D	–	Three Dimensional
DOF	–	Degrees of Freedom
HCI	–	Human Computer Interaction
GUI	–	Graphical User Interface

CHAPTER 1: INTRODUCTION

1.1 Background of Study

In recent years, the integration of robotics in the healthcare sector has significantly transformed the field of surgery. Robotic-assisted surgical systems provide enhanced precision, flexibility, and control compared to traditional manual surgical procedures. These systems reduce human error, improve accuracy, and enable minimally invasive operations, leading to faster patient recovery and reduced risk of complications.

With the advancement of computer graphics and simulation technologies, it has become possible to model complex surgical environments virtually. Simulation plays a crucial role in training, research, and development of robotic systems without involving real patients. MATLAB, with its powerful visualization and computational capabilities, provides an effective platform for developing such simulations.

This project focuses on designing a 3D simulation of a surgical robot operating room, where multiple robotic arms interact with a patient model. The study combines concepts of robotics, computer graphics, and human-machine interaction to create a realistic and interactive environment.

1.2 Problem Statement

Traditional surgical procedures heavily depend on the skill and precision of surgeons, which can sometimes lead to errors due to fatigue, limited visibility, or hand tremors. While robotic-assisted surgery exists, understanding and developing such systems require safe and controlled environments for testing and training.

There is a need for a simulation-based system that can:

- Represent a realistic surgical environment
- Demonstrate the working of multiple robotic arms
- Allow user interaction for controlling robotic movements
- Provide a platform for experimenting with robotic behaviour without real-world risks

This project aims to address these challenges by developing a MATLAB-based simulation of a surgical robot system with interactive controls and visualization.

1.3 Motivation

The motivation behind this project comes from the increasing demand for automation and precision in the medical field. Surgical robots are becoming an essential part of modern healthcare systems, and understanding their working is crucial for future engineers.

Key motivating factors include:

- Growing importance of robotic-assisted surgery
- Need for safe training platforms without real patients
- Interest in combining programming, simulation, and real-world applications
- Opportunity to explore interdisciplinary concepts such as robotics, AI, and visualization

This project provides a foundation for learning and experimenting with robotic systems in a controlled and risk-free environment.

1.4 Scope of the Project

The scope of this project is focused on developing a virtual simulation of a surgical robot operating room using MATLAB. The system includes multiple robotic arms, a patient model, and an operation table within a 3D environment.

The main scope includes:

- Designing a realistic 3D operating room environment
- Modelling multiple robotic arms with movement and rotation
- Implementing keyboard-based controls for interaction
- Visualizing surgical procedures in a simulated environment

However, the project has certain limitations:

- It does not include real-time medical data or actual robotic hardware
- Artificial Intelligence and automation are not fully implemented
- Collision detection and advanced path planning are limited

Despite these limitations, the project serves as a strong base for further development in robotic surgery systems.

1.5 Organization of Report

This report is organized into multiple chapters to provide a structured understanding of the project:

- **Chapter 1: Introduction**
Presents the background, problem statement, motivation, scope, and overall structure of the project.
- **Chapter 2: Literature Review and Requirement Analysis**
Discusses existing systems, research work, and technologies used in robotic surgery.
- **Chapter 3: Methodology and System Design**
Explains the system architecture, design approach, algorithms, and tools used in the project.
- **Chapter 4: Implementation and Testing**
Describes the coding process, module development, and testing of the system.
- **Chapter 5: Results and Discussion**
Presents the output of the simulation along with analysis and observations.
- **Chapter 6: Conclusion and Future Scope**
Summarizes the project and suggests possible improvements and future enhancements.

CHAPTER 2:LITERATURE REVIEW/REQUIREMENTS ANALYSIS AND SPECIFICATIONS

2.1 Summary of Existing Work

Robotic-assisted surgery has evolved significantly over the past few decades, with systems like the *da Vinci Surgical System* revolutionizing minimally invasive procedures. These systems allow surgeons to perform complex operations with enhanced precision, flexibility, and control through robotic arms and advanced visualization.

Existing research and systems in this domain focus on:

- High-precision robotic arms for surgical procedures
- 3D visualization systems for better depth perception
- Remote surgery (teleoperation) using network-based control
- Integration of AI and machine learning for decision support

Simulation-based approaches are also widely used for:

- Training surgeons without real patients
- Testing robotic movements in a controlled environment
- Visualizing surgical processes in 3D

Many researchers have used tools such as MATLAB, ROS (Robot Operating System), and Unity for creating robotic simulations. MATLAB, in particular, is widely used due to its strong capabilities in mathematical modelling and 3D visualization.

2.2 Research Gaps

Despite advancements in robotic surgery systems, several limitations and gaps still exist, especially in simulation-based projects:

- Most real-world systems are expensive and not easily accessible for students and beginners
- Many simulations focus only on single robotic arms, whereas real surgeries may involve multiple robots
- Limited emphasis on interactive control systems for user-based learning
- Lack of simple and understandable models for educational purposes
- Advanced features like collision detection, path planning, and AI-based automation are often missing or complex

In this project, an attempt has been made to:

- Develop a multi-robot simulation system
- Provide interactive control using keyboard inputs
- Create a simple yet realistic 3D surgical environment
- Lay a foundation for future integration of AI and automation

2.3 Comparative Analysis

A comparison between existing systems and the proposed project is shown below:

Feature	Existing Systems	Proposed Project
Cost	Very high (real robotic systems)	Low (simulation-based)
Accessibility	Limited	Easily accessible (MATLAB-based)
Number of Robots	Usually single or complex multi-robot systems	Multiple robots (4 robots)
User Interaction	Limited (complex controls)	Simple keyboard-based control
Visualization	Highly advanced	Moderate but realistic
AI Integration	Present in advanced systems	Not implemented (future scope)
Training Capability	Professional-level	Educational-level

2.4 Requirement Analysis

Functional Requirements

The system should:

- Display a 3D surgical operating room environment
- Include a patient model and operation table
- Simulate multiple robotic arms
- Allow user interaction through keyboard controls
- Enable movement, rotation, and height adjustment of robots

Non-Functional Requirements

The system should:

- Be easy to use and understand
- Provide smooth graphical visualization

- Be lightweight and efficient
- Run on standard systems without high hardware requirements

2.5 Tools and Technologies Used

- MATLAB → for simulation and visualization
- 3D plotting functions (surf, plot3, meshgrid)
- Mathematical functions → trigonometry for movement
- Event handling → keyboard input control

2.6 System Specifications

Hardware Requirements

- Processor: Intel i3 or above
- RAM: 4 GB or higher
- Storage: Minimum 2 GB free space

Software Requirements

- MATLAB (R2018 or later recommended)
- Operating System: Windows / Linux / Mac

CHAPTER 3: METHODOLOGY / SYSTEM DESIGN

3.1 System Overview

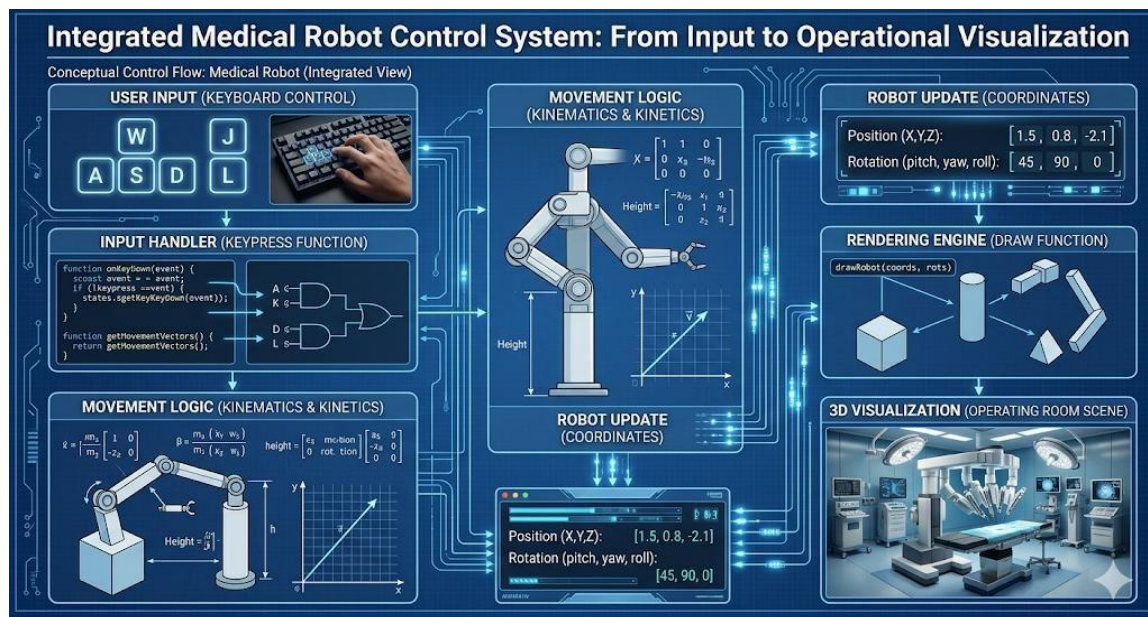
The proposed system is a 3D simulation of a surgical robot operating room developed using MATLAB. The system models multiple robotic arms interacting with a patient placed on an operation table. The simulation allows real-time user interaction through keyboard inputs to control the movement, rotation, and positioning of robotic arms.

The methodology involves:

- Creating a 3D environment using MATLAB graphical functions
- Designing robotic arms and patient structures using geometric shapes
- Implementing control logic for robot movement
- Updating the simulation dynamically based on user inputs

3.2 System Architecture / Block Diagram

Below is the conceptual architecture of the system:

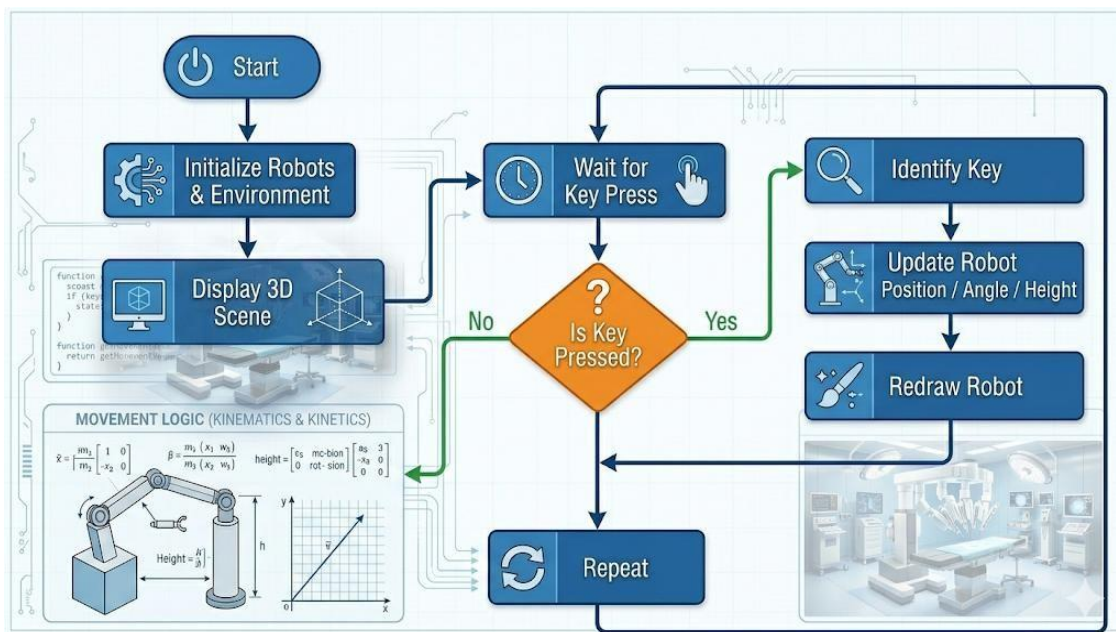


Explanation:

- User keyboard input triggers movement
- Logic calculates new positions
- Robots are updated and redrawn
- Final output is displayed in 3D

3.3 Flowchart

Working Flow of the System



This loop continues until the program is closed.

3.4 Algorithms Used

Algorithm 1: Robot Movement (Linear Movement)

1. Start
2. Get current robot position (x, y)
3. Calculate distance from center (operation table)
4. Normalize direction vector (unit vector)

5. If key = 'W' → move towards center
6. If key = 'S' → move away from center
7. Update position
8. Redraw robot
9. End

Algorithm 2: Circular Movement

1. Start
2. Calculate angle using $\text{atan2}(\text{dy}, \text{dx})$
3. If key = 'A' → increase angle
4. If key = 'D' → decrease angle
5. Compute new position using:
 - $x = \text{center } X + r * \cos(\text{angle})$
 - $y = \text{center } Y + r * \sin(\text{angle})$
6. Update robot position
7. Redraw robot
8. End

Algorithm 3: Arm Control

1. Start
2. If key = 'J' → increase arm angle
3. If key = 'L' → decrease arm angle
4. If key = 'N' → increase height
5. If key = 'M' → decrease height
6. Update robot structure
7. Redraw robot
8. End

3.5 Dataset Details (If Applicable)

This project does not use any external dataset.

Instead, it relies on:

- Mathematical modelling
- Predefined geometric shapes (cylinder, sphere)
- Coordinate-based positioning

The simulation is entirely generated using:

- MATLAB built-in functions
- Real-time calculations for movement and rendering

3.6 Hardware and Software Requirements

Hardware Requirements

- Processor: Intel i3 or higher
- RAM: Minimum 4 GB
- Graphics support for 3D rendering
- Keyboard for interaction

Software Requirements

- MATLAB (R2018 or later recommended)
- Operating System: Windows / Linux / macOS

3.7 Design Considerations

- The system is designed for simplicity and visualization
- Focus on real-time interaction
- Modular design using functions like:
 - Draw Robot()
 - move Robot()
- Efficient use of MATLAB graphics

CHAPTER 4: IMPLEMENTATION

4.1 Step-by-Step Implementation Process

The implementation of the surgical robot simulation system is carried out in MATLAB using a structured approach. The complete process is described below:

Step 1: Initialization

- MATLAB environment is cleared using:
 - `clc`, `clear`, `close all`
- Global variables (robots, handles) are initialized
- Initial positions, angles, and heights of robots are defined Four robots are placed at fixed coordinates around the operation table.

Step 2: Creating the 3D Environment

- A 3D figure window is created using:
 - `figure()`
- Axes, grid, and labels are configured
- Floor is created using `meshgrid()` and `surf()`

This forms the base of the operating room.

Step 3: Modelling the Operation Table

- The base of the table is created using a cylindrical shape
- The top surface is created using a rectangular mesh

Positioned at the center coordinates (25, 25)

Step 4: Patient Modelling

The patient is designed using multiple geometric components:

- Torso → Cylinder
- Shoulders → Surface curve
- Neck → Small cylinder
- Head → Sphere
- Pillow → Flat surface

- Surgical area → Highlighted region

All parts are aligned properly using coordinate transformations.

Step 5: Robot Design

Each robot consists of:

- Base → Cylinder
- Vertical arm → Cylinder
- Movable arm → Line (plot3)
- Surgical tool (needle) → Cylinder + cone

The function draw Robot() is used to create robots dynamically.

Step 6: Rendering Robots

- Robots are stored in a structure array
- A loop is used to draw all robots
- Handles are saved for updating later

Step 7: Implementing User Interaction

- Keyboard control is enabled using:

`figure('Key Press Fcn', @moveRobot)`

- The function move Robot() handles key inputs

Step 8: Movement Logic

Based on key press:

- W / S → Move towards / away from center
- A / D → Circular movement
- J / L → Rotate arm
- N / M → Adjust height

Position and angle are recalculated using trigonometric functions.

Step 9: Updating Simulation

- Old robot graphics are deleted
- New updated robot is redrawn
- Draw now ensures real-time update

4.2 Modules Description

The system is divided into the following modules:

Module 1: Environment Setup

- Creates 3D space
- Defines axes, grid, and lighting

Module 2: Patient Model

- Builds human body structure
- Uses surface and geometric shapes

Module 3: Robot Model

- Defines robot structure
- Includes base, arm, and surgical tool

Module 4: Control Module

- Handles keyboard inputs
- Controls movement and rotation

Module 5: Rendering Module

- Updates robot positions
- Handles real-time visualization

4.3 Screenshots (Software Output)

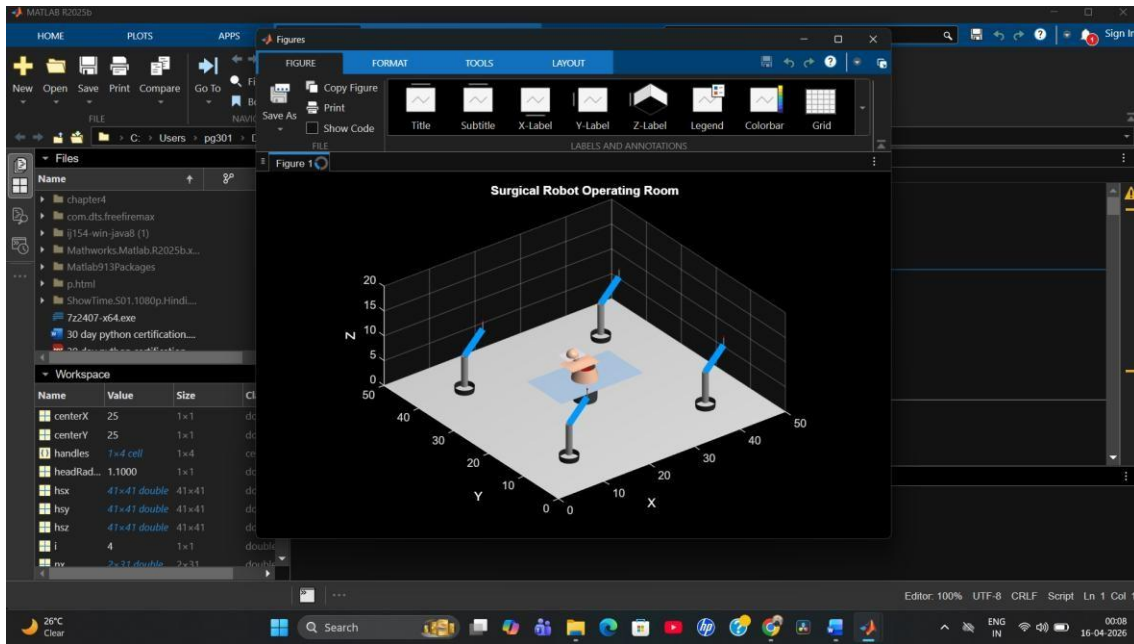


Figure 4.3.1: 3D Surgical Robot Operating Room Environment

The above figure shows the initial 3D simulation setup of the surgical robot operating room developed using MATLAB. It includes a patient model placed at the center of the operation table, surrounded by four robotic arms positioned at different locations. The operating room environment is visualized using a 3D coordinate system with proper lighting and surface rendering.

Each robotic arm is designed to perform surgical operations and can be controlled using keyboard inputs. The figure demonstrates the basic structure and arrangement of the system before any movement or interaction takes place. This setup serves as the foundation for further simulation, where robots can move, rotate, and adjust their tools in real time.

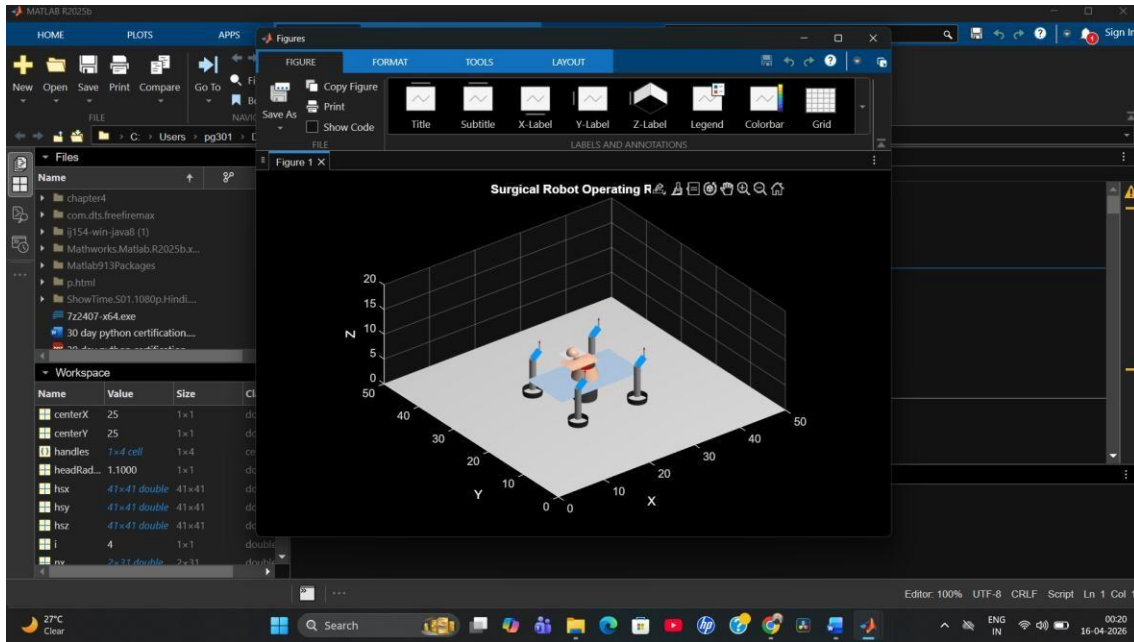


Figure 4.3.2: 3D Surgical Robot Operating Room with Robots Positioned Around Patient

The above figure represents the 3D simulation of a surgical robot operating room developed using MATLAB. It shows a patient placed on the operation table at the center, surrounded by four robotic arms positioned at different locations. The environment is visualized using a three-dimensional coordinate system, providing a clear view of the spatial arrangement.

Each robotic arm is designed to perform surgical tasks and is equipped with a movable arm and tool. The robots are symmetrically placed around the patient to simulate a real surgical setup. This figure demonstrates the initial configuration of the system before any movement or interaction is applied. The setup serves as the base for further operations such as robot movement, rotation, and tool adjustment using user inputs.

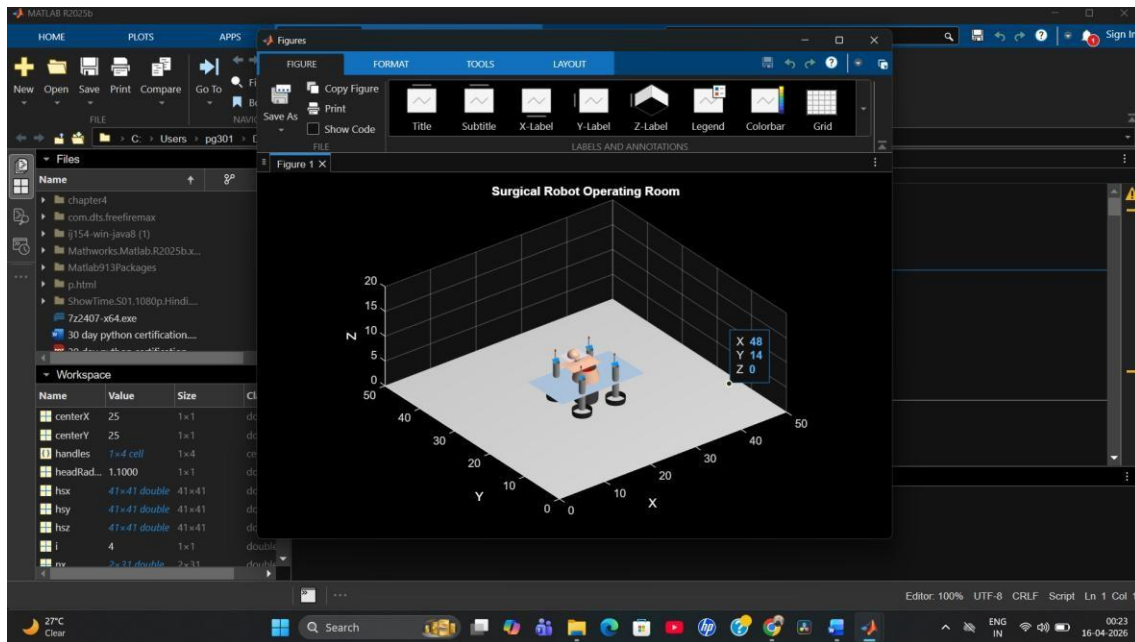


Figure 4.3.3: Robotic Arms Moving Towards the Surgical Area

The above figure shows the movement of robotic arms towards the patient in the surgical robot operating room simulation. Compared to the initial setup, the robots have moved closer to the surgical area, demonstrating their ability to perform coordinated actions around the patient.

The robotic arms are controlled using keyboard inputs, allowing them to adjust their position, angle, and height in real time. This figure highlights the dynamic behaviour of the system, where robots can approach the patient and prepare for surgical operations. The simulation effectively demonstrates real-time interaction and control of multiple robotic arms in a 3D environment.

4.4 Hardware Setup Images

This project is purely software-based and does not involve physical hardware. However, the system requires:

- A computer/laptop
- MATLAB software installed
- Keyboard for user interaction

Therefore, no hardware setup images are required.

4.5 Implementation Challenges

During development, the following challenges were encountered:

- Managing real-time updates without lag
- Aligning patient model correctly on the table
- Handling multiple robots simultaneously
- Implementing smooth movement using trigonometry

4.6 Improvements Implemented

- Optimized robot drawing functions
- Added lighting and shading for realism
- Improved alignment of patient model
- Enabled smooth keyboard interaction

CHAPTER 5: RESULTS AND DISCUSSION

5.1 Output Results

The developed system successfully simulates a 3D surgical robot operating room environment using MATLAB. The simulation includes a patient model placed on an operation table and four robotic arms positioned around the patient.

The following outputs were observed:

- A realistic 3D operating room setup with floor, table, and patient
- Four robotic arms positioned symmetrically around the patient
- Smooth real-time movement of robots using keyboard controls
- Adjustable robot arm rotation and tool height
- Visualization of a surgical area on the patient

The system responds dynamically to user inputs, updating robot positions and orientations in real time without requiring program restart.

5.2 Graphs / Charts / Evaluations

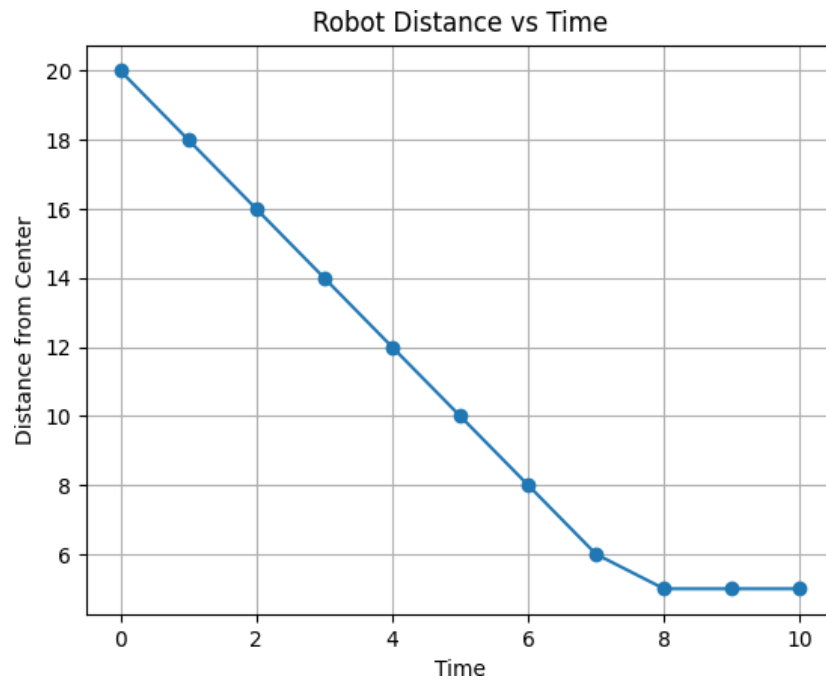
Since the project is simulation-based, graphical evaluation is mainly visual rather than numerical. However, the system can be evaluated based on performance and responsiveness:

Performance Evaluation Parameters

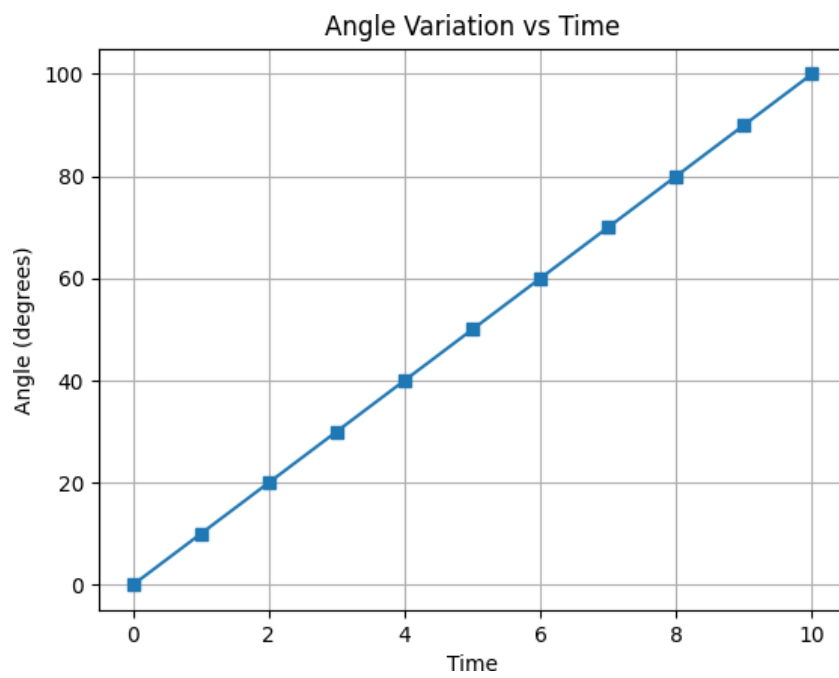
Parameter	Observation
Response Time	Immediate response to key input
Rendering Speed	Smooth visualization with minimal lag
Accuracy of Movement	Precise circular and linear motion
Stability	No crashes during execution

Possible Graphs (Optional for Report)

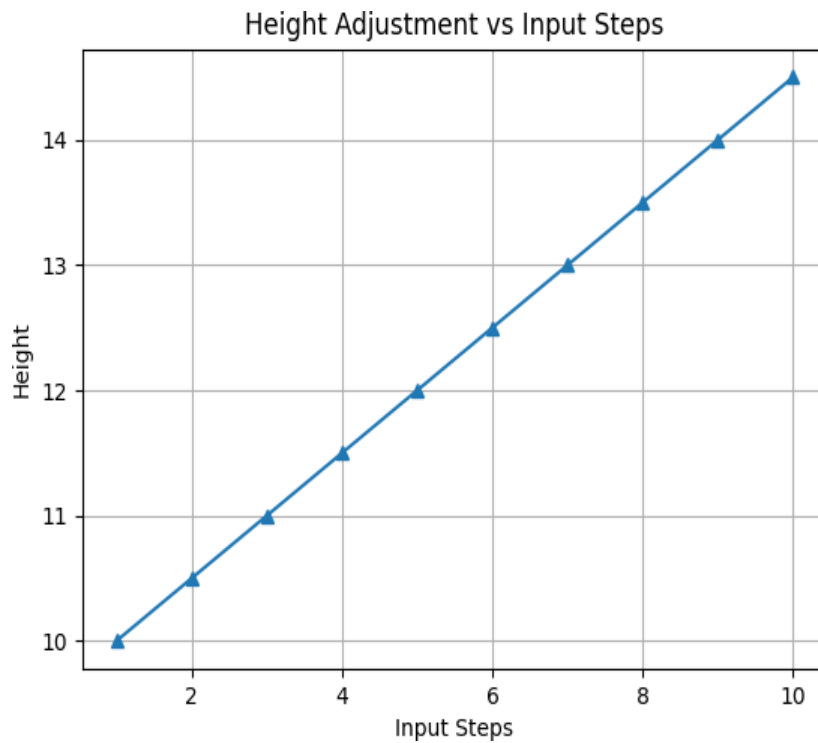
- Robot distance vs time



- Angle variation vs time



- Height adjustment vs input steps



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5.3 Comparative Analysis

The performance of the proposed system is compared with traditional and advanced robotic systems:

Feature	Traditional Surgery	Advanced Robotic Systems	Proposed System
Precision	Moderate	Very High	Moderate
Cost	Low	Very High	Very Low
Accessibility	High	Limited	High
Automation	None	High (AI-based)	Manual (user-controlled)
Visualization	Limited	Advanced 3D	3D Simulation
Training Capability	Limited	Professional	Educational

Observation:

- Proposed system is cost-effective and educational

- But lacks AI and real-world integration

5.4 Interpretation of Results

The results indicate that the developed system effectively demonstrates the concept of robotic-assisted surgery in a virtual environment.

Key interpretations:

- The system successfully simulates multi-robot coordination around a patient
- Real-time keyboard interaction provides user control and flexibility
- The use of 3D graphics enhances visual understanding of robotic movement
- The project is suitable for educational and demonstration purposes

However, some limitations were observed:

- Lack of automatic decision-making (AI)
- No collision detection mechanism
- Movement is manual rather than autonomous

Despite these limitations, the system provides a strong foundation for future improvements such as AI integration, path planning, and real-world robotic implementation.

CHAPTER 6: CONCLUSION & FUTURE SCOPE

6.1 Summary of the Project

This project presents the design and development of a 3D simulation of a surgical robot operating room using MATLAB. The system models a realistic environment that includes a patient placed on an operation table and multiple robotic arms positioned around the patient.

The simulation enables real-time interaction through keyboard controls, allowing users to move robotic arms, adjust their angles, and control the height of surgical tools. The use of MATLAB's graphical functions helps in creating a visually interactive and dynamic environment.

Overall, the project successfully demonstrates the working concept of robotic-assisted surgery in a virtual setup, making it useful for educational and demonstration purposes.

6.2 Key Findings

The following key findings were observed during the development and testing of the system:

- The system effectively simulates a multi-robot surgical environment
- Real-time keyboard interaction allows smooth and responsive control
- MATLAB provides a powerful platform for 3D visualization and simulation
- The project helps in understanding the basic principles of robotic movement and coordination
- The simulation can serve as a training and learning tool for students

6.3 Limitations

Despite achieving its objectives, the project has certain limitations:

- The system does not include Artificial Intelligence or automation
- All robot movements are manually controlled by the user
- No collision detection mechanism is implemented
- The simulation does not use real medical data or hardware integration
- The graphical model, although realistic, is still simplified compared to real surgical systems

6.4 Future Enhancement Ideas

The project can be further improved and extended in the following ways:

- **Integration of Artificial Intelligence (AI):**
Implement automated decision-making for robotic movement and surgical assistance
- **Collision Detection System:**
Add algorithms to prevent robots from colliding with each other or the patient
- **Path Planning Algorithms:**
Enable robots to move automatically using shortest and safest paths
- **Real-Time Medical Data Integration:**
Connect the system with imaging techniques such as MRI or CT scans
- **Hardware Implementation:**
Interface the simulation with real robotic arms for practical applications
- **Enhanced Graphics and UI:**
Improve visualization using advanced tools or game engines
- **Remote Control (Teleoperation):**
Allow control of robots over a network for remote surgery simulation

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